01211373 Machine Learning and Programming for Industry 2026 (IUP)

A 14-Week Undergraduate Course (3rd Year) — Python / scikit-learn / PyTorch

Main text: Raschka, Liu & Mirjalili, *Machine Learning with PyTorch and Scikit-Learn* (Packt, 2022). Chapter references (Ch. #, page #) below refer to this book throughout.

Course Philosophy

Every topic is chosen because it has a direct application in robotics or industrial automation: sensor calibration, fault detection, predictive maintenance, visual inspection, time-series prediction, and learning-based control. The course follows the textbook's chapter order closely so students can read ahead, but **trims or compresses topics with limited use in current practice** — bag-of-words NLP, full derivations of historical algorithms, AdaBoost, LDA/t-SNE, GANs, and GNNs are either dropped, folded into a single lecture as a brief pointer, or left as optional reading, freeing time for the material students will actually reach for: gradient boosting, PyTorch fundamentals, CNNs, LSTMs, attention/transformers, and reinforcement learning.

Prerequisites: Python fundamentals, linear algebra, probability/statistics, basic control systems.

Weekly format: 1 lecture (theory + robotics framing) + 1 lab (hands-on implementation).

Weekly Schedule

Part 1 — Classical Machine Learning (scikit-learn)

Week	Topic	Book Chapters	Learning Outcomes	Lab	Robotics / Automation Application
1	ML Foundations & Tooling	Ch. 1 (p.1); Ch. 2 (p.19) — brief recap only	ML taxonomy (supervised/unsupervised/RL); perceptron & Adaline *concept* (not full derivation); Python/NumPy/Pandas refresher; scikit-learn & PyTorch environment setup	Load and visualize real sensor datasets (IMU, force/torque, vibration); train a first perceptron/logistic model	Framing ML as a complement to state estimation and control
2	Classifiers Tour & Data Preprocessing	Ch. 3 (p.53); Ch. 4 (p.105)	Logistic regression, SVM, decision trees, k-NN; handling missing/categorical data; feature scaling	Classify machine operating states (normal/fault) from sensor features	Multiclass fault detection in industrial equipment

Week	Topic	Book Chapters	Learning Outcomes	Lab	Robotics / Automation Application
3	Regression Analysis	Ch. 9 (p.269)	OLS, regularized regression (Ridge/Lasso), polynomial regression, random forest regression	Fit a regression model to calibrate a force sensor / estimate joint friction	Sensor calibration, static system identification
4	Ensemble Learning	Ch. 7 (p.205) — emphasis on §"Gradient boosting" and XGBoost; AdaBoost covered briefly as historical context only	Bagging, random forests, gradient boosting, XGBoost	Predict remaining-useful-life risk class from vibration/temperature features	Predictive maintenance, root-cause feature importance
5	Dimensionality Reduction & Clustering	Ch. 5 (p.139) — PCA is the focus, LDA/t-SNE mentioned only briefly; Ch. 10 (p.305)	PCA; k-means, hierarchical clustering, DBSCAN	Cluster machine states from unlabeled vibration spectra; anomaly flagging	Unsupervised anomaly detection when fault labels are unavailable
6	Model Evaluation & Hyperparameter Tuning	Ch. 6 (p.171)	Pipelines, k-fold cross-validation, learning/validation curves, grid/randomized search, confusion matrix, ROC	**Project Checkpoint 1**: deliver a validated classical ML pipeline on a real dataset	Rigorous evaluation for safety-relevant automation decisions

Part 2 — Deep Learning (PyTorch)

Week	Topic	Book Chapters	Learning Outcomes	Lab	Robotics / Automation Application
7	From Neural Net Foundations to PyTorch	Ch. 11 (p.335) — condensed to core forward/backprop intuition, not the full from-scratch build; Ch. 12 (p.369)	MLP architecture and backpropagation (conceptual); PyTorch tensors, autograd, DataLoader; building/training a first PyTorch model	Re-implement the Week 2 classifier as a PyTorch MLP	Comparing classical vs. neural models on the same fault-detection task
8	PyTorch Mechanics	Ch. 13 (p.409)	Computation graphs, `nn.Module`, custom layers, loss functions, saving/loading models	Regression project: predict a continuous physical quantity (e.g., robot joint torque / friction residual), following the book's "fuel efficiency" project pattern	Learning-based approximation of nonlinear robot dynamics
9	Convolutional Neural Networks	Ch. 14 (p.451)	Convolution/pooling, CNN architecture, regularization (dropout, batch norm), building a CNN in `torch.nn`	Train a CNN for visual defect detection on an industrial image set	Automated visual quality inspection on a production line
10	Recurrent Networks & Sequential Data	Ch. 15 (p.499)	RNN dataflow, vanishing gradients, LSTM cells; building an RNN/LSTM in PyTorch	Forecast a time-series signal (e.g., tool wear, temperature trend) with an LSTM	Predictive maintenance and trajectory/time-series forecasting
11	Attention & Transformers (concept-level)	Ch. 16 (p.539) — self-attention and the transformer architecture only; BERT/GPT fine-tuning details (§"Fine-tuning a	Why attention outperforms plain RNNs on long sequences; scaled dot-product attention; where transformer-based perception	Implement scaled dot-product attention from scratch; inspect a pretrained	Multi-sensor fusion and modern perception backbones

Week	Topic	Book Chapters	Learning Outcomes	Lab	Robotics / Automation Application
		BERT model") skipped as NLP-specific	models (e.g., vision transformers, sensor fusion) fit in robotics	vision transformer	
12	Reinforcement Learning	Ch. 19 (p.673)	MDPs, value functions, Q-learning, deep Q-learning (DQN)	Solve a grid-world task with Q-learning; extend toward a simple simulated control task	Learning-based control policies for robotic tasks
13	Deployment & Sim-to-Real	*(synthesis — not a dedicated book chapter)*; brief pointer to Ch. 17 (p.589, GANs) for synthetic training-data generation; brief pointer to Ch. 18 (p.637, GNNs) as further reading for graph-structured problems (e.g., multi-robot systems, kinematic-chain models)	Sim-to-real gap, model export/quantization, latency constraints, inference on embedded hardware, basic ROS integration pattern	Export and benchmark a trained model for real-time inference	Deploying ML models on real robot/PLC-class hardware
14	Responsible ML & Summary	Ch. 19 summary (p.714) as closing framing	Explainability, failure modes, safety considerations, monitoring in production	-	Reflecting on when/why to trust an ML model in a safety-relevant loop

What Was Cut or Compressed, and Why

- **Ch. 8 (Sentiment Analysis / bag-of-words NLP)** — dropped entirely. It's the book's only fully text-focused chapter, has no natural robotics/automation tie-in, and the bag-of-words approach it teaches has been largely superseded by the transformer-based methods covered (at concept level) in Week 11.
- **Ch. 2 (Perceptron/Adaline from scratch)** — compressed to a short conceptual recap in Week 1 rather than a full derivation-and-implementation pass, since Week 7–8 rebuild the same ideas properly in PyTorch.
- **AdaBoost (part of Ch. 7)** — mentioned as historical context only; gradient boosting and XGBoost get the lab time, since they're what students will actually encounter in predictive-maintenance pipelines today.

- **LDA and t-SNE (part of Ch. 5)** — mentioned briefly; PCA carries the week since it's the dimensionality-reduction tool most used in practice for sensor/vibration data.
- **Ch. 17 (GANs)** and **Ch. 18 (Graph Neural Networks)** — no dedicated weeks. GANs get a one-slide pointer in Week 13 (useful for synthetic defect-image generation when labeled data is scarce); GNNs get a pointer as further reading for students who go on to multi-robot or kinematic-graph problems. Both are interesting but low-yield for a one-semester survey with this program's focus.

This frees roughly 1.5 weeks of contact time compared to following the book chapter-for-chapter, which is why the schedule still closes with a full deployment week (13) and capstone week (14) — the two most directly useful weeks for this audience, and easy to lose if the syllabus just mirrors the table of contents.

Semester Project (runs Weeks 2–14)

Students select one problem, preferrably tied to a real or simulated robotics/automation dataset:

- Fault detection on a rotating machine (bearing/vibration dataset)
- Visual inspection of manufactured parts
- Predictive maintenance from multivariate sensor time series
- Learning a residual dynamics model for a robotic arm
- A simple RL-based controller in simulation
- Related topics

Submit the project report with video at the end of this course (deadline TBD).

Assessment

Component	Weight
Lab assignments (weekly)	30%
Class project	30%
Midterm exam (covers Weeks 1–7 concepts)	20%
Final exam (covers Weeks 8–14 concepts)	20%

Tools & Environment

- **Language:** Python 3.x
- **Core libraries:** NumPy, Pandas, Matplotlib
- **Classical ML:** scikit-learn
- **Deep learning:** PyTorch (+ torchvision for the CNN week)
- **Optional:** Gymnasium (formerly OpenAI Gym) for the RL week, matching the book's Ch. 19 examples; ONNX or TorchScript for the deployment week
- **Suggested datasets:** CWRU Bearing Dataset (fault detection), NASA C-MAPSS (predictive maintenance/RUL), a small custom or public defect-inspection image set, IMU/force-torque logs (can be simulated if lab hardware is unavailable)

References

- **Primary text:** Raschka, Liu & Mirjalili, *Machine Learning with PyTorch and Scikit-Learn* (Packt, 2022) — used throughout, chapter mapping above.
- Sutton & Barto, *Reinforcement Learning: An Introduction* — supplementary depth for Week 12, pairs directly with the book's Ch. 19 treatment.
- Selected papers on learning-based control and sim-to-real transfer, assigned as short readings in Weeks 12–13.